

Problem 1 : Basic Exam, Fall 2011 [TODO]

Let (X, d) be a compact metric space and let $f : X \rightarrow X$ be a map satisfying

$$d(f(x), f(y)) < d(x, y), \quad \forall x, y \in X, x \neq y.$$

Prove that there is a unique point $x \in X$ such that $f(x) = x$.

Solution. Look at $g : X \rightarrow \mathbb{R}$ where $g(x) = d(f(x), x)$. X is compact and thus $\exists z \in X$ such that $g(x) \leq g(y)$ for all $y \in X$. Assume the contrary that $f(z) \neq z$. Consider $g(f(z)) = d(f(f(z)), f(z)) < d(f(z), z) = g(z)$ which is a contradiction.

Uniqueness is straight forward because x, y fixed points then $d(f(x), f(y)) = d(x, y)$ which contradicts the assumption.

Problem 2 : Basic Exam, Fall 2020 [TODO]

Find all positive values of x for which the series

$$\sum_{n=0}^{\infty} \frac{(xn)^n}{n!}$$

converges.

Solution.

$$H = \limsup_{n \rightarrow \infty} \frac{n}{(n!)^{\frac{1}{n}}}$$

We have Sterling's Approximation, that $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$.

Problem 3 : Basic Exam, Spring 2018 [TODO]

(a) Let $U \subseteq \mathbb{R}^n$ be non-empty, open, and connected. Suppose that $f : U \rightarrow \mathbb{R}$ is such that all the first order partial derivatives of f exist and vanish at each point of U . Prove that f is constant in U . [TODO]

(b) Let $\Omega \subseteq \mathbb{R}^n$ be open non-empty and let $f : \Omega \rightarrow \mathbb{R}$ be such that all the first order partial derivatives of f exist and are bounded in Ω . Prove that f is continuous in Ω . [TODO]

Solution. (a): $\forall x \in U$, there exists a box neighborhood $B_x \ni x$. By Taylor's Theorem we have,

$$f\left(x + \sum_{i=1}^{k-1} s_i e_i + t e_k\right) = g_k(t; \vec{s}) = f\left(x + \sum_{i=1}^{k-1} s_i e_i + t e_k\right) + (\partial_k f)\left(x + \sum_{i=1}^{k-1} s_i e_i\right) t.$$

For $y \in B_x$. There exists a $\delta_1, \dots, \delta_n$ such that

$$y - x = \begin{pmatrix} \delta_1 \\ \vdots \\ \delta_n \end{pmatrix}.$$

$$|f(y) - f(x)| \leq \sum_{i=1}^n |g_{i-1}(\delta_i, \vec{\delta}) - g_i(\delta_{i+1}, (\delta))| = 0$$

Thus we have constant value.

(b) We repeat a similar computation to the first thing.

Problem 4 : Basic Exam, Fall 2025 [TODO]

Let $f : [0, \infty) \rightarrow [0, \infty)$ be continuous on $[0, \infty)$ and continuously differentiable on $(0, \infty)$ with $\int_0^\infty |f'(x)| dx < \infty$. Prove that the limit

$$\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n f(k) - \int_0^n f(x) dx \right)$$

exists.

Solution.

$$\begin{aligned} \sum_{k=1}^n f(k) - \int_0^n f(x) dx &= \sum_{k=1}^n \int_{k-1}^k (f(k) - f(x)) dx \\ &= \sum_{k=1}^n \int_0^1 (f(k) - f(x + k - 1)) dx \\ &= \sum_{k=1}^n \int_0^1 \int_{x+k-1}^k f'(t) dt dx \\ &= \sum_{k=1}^n \int_{k-1}^k (t - k + 1) f'(t) dt \\ &= \sum_{k=1}^n \int_{k-1}^k (t - \lfloor t \rfloor) f'(t) dt \\ &= \int_0^n (t - \lfloor t \rfloor) f'(t) dt \\ &\leq \int_0^n |f'(t)| dt \end{aligned}$$

By the Direct comparison test, the limit given converges